

TShark Commands

◆ `tshark -v`

☞ Shows the **TShark version** and basic information.

Use:

Check if TShark is installed and which version you have.

◆ `tshark -D`

☞ Shows **all network interfaces** available for capture.

Use:

Helps you choose the correct interface (`eth0`, `wlan0`, etc.).

◆ `tshark -i eth0`

☞ Starts capturing packets on the **eth0 (wired) interface**.

Use:

Sniff network traffic on a LAN cable.

◆ `tshark -i eth0 -c 10`

☞ Captures **only 10 packets** on the **eth0 (wired) interface**.

Use:

Quick test capture without flooding the screen.

🔒 Sniffing Specific Protocols

◆ `sudo tshark -i wlan0 udp`

☞ Captures **only UDP traffic**.

◆ `sudo tshark -i wlan0 icmp`

☞ Captures **ping packets**.

◆ `sudo tshark -i wlan0 tcp`

☞ Captures **TCP traffic**.

★ **Use:**

Analyze only the protocol you want.

◆ `tshark -i eth0 host <IP>`

☞ Captures traffic **to or from a specific IP address**.

Use:

Monitor one target system.

◆ `sudo tshark -i wlan0 -x -c 2 host <Kali IP>`

☞ Captures **2 packets** from/to Kali IP and shows **hex + ASCII data**.

Use:

See raw packet data for analysis.

📁 Working with Capture Files

◆ `sudo tshark -w /tmp/nlog.pcap -i wlan0 host <Kali IP>`

☞ Captures traffic and **saves it to a file**.

Use:

Save packets for later analysis.

◆ **ls -l /tmp/nlog.pcap**

☞ Confirms the file exists and shows **file size & permissions**.

◆ **file /tmp/nlog.pcap**

☞ Tells the **type of file** (pcap capture file).

◆ **sudo tshark -r /tmp/nlog.pcap**

☞ **Reads and displays packets** from the saved file.

Use:

Analyze old traffic without capturing again.

Super Simple Summary

Command	What it Does
-v	Shows version
-D	Lists interfaces
-i	Selects interface
-c	Limits packets
udp / tcp / icmp	Filters protocol
host <IP>	Filters IP
-x	Shows hex data
-w	Saves capture
-r	Reads capture

Display Filters

1 How to Use Display Filters in TShark

◆ Syntax

```
tshark -r capture.pcap -Y <display-filter>
```

✦ -r → read capture file

✦ -Y → apply display filter

✓ Read the FULL packet details (Best next step)

-v (verbose)

```
tshark -r httpres.pcap -Y 'frame contains "update"' -V (Capital)
```

```
tshark -i eth0 -Y 'frame contains "pass"' -V
```

◆ Example 1: Show Only HTTP Traffic

```
tshark -i eth0 -Y http
```

☞ Shows only website (HTTP) packets.

◆ Example 2: Show Only DNS Traffic

```
tshark -i eth0 -Y dns
```

☞ Shows domain name queries.

◆ Example 3: Show Traffic of One IP

```
tshark -i eth0 -Y "ip.addr == 192.168.1.10"
```

☞ Shows all packets to/from that IP.

◆ Example 4: Show Only SSH Packets

```
tshark -i eth0 -Y "tcp.port == 22"
```

☞ Shows SSH traffic.

◆ Example 5: Find Login (POST) Requests

```
tshark -i eth0 -Y "http.request.method == POST"
```

☞ Used in CTFs & pentesting.

2□ Display Filters While Live Capture

You can also apply filters **during live capture** (view only):

```
sudo tshark -i wlan0 -Y dns
```

☞ Captures everything but **shows only DNS packets**.

3□ Common Display Filters (Must Know 🔥)

Filter	What it Shows
http	HTTP traffic
dns	DNS queries
tcp	TCP packets
udp	UDP packets
icmp	Ping packets
ssh	SSH traffic

Filter	What it Shows
<code>tcp.port == 80</code>	HTTP
<code>tcp.port == 22</code>	SSH
<code>ip.src == x.x.x.x</code>	Source IP
<code>ip.dst == x.x.x.x</code>	Destination IP
<code>frame contains "password"</code>	Password text (HTTP)

✓ What are Logical Filters in TShark?

Logical filters let you **combine conditions** so you can see **exactly the packets you want**. They are used with **display filters** (–Y), not capture filters.

☞ Same logic as Wireshark, but used in the **terminal**.

🔑 Logical Operators Used in TShark

TShark uses **3 logical operators**:

Operator	Meaning	Simple Words
<code>&&</code>	AND	both must be true
<code>,</code>		
<code>!</code>	NOT	exclude something

Logical Filters

1. AND Operator (&&)

☞ **Meaning: Both conditions must be true**

◆ Example 1: HTTP traffic from one IP

```
tshark -i eth0 -Y "ip.addr == 192.168.1.10 && http"
```

✓ Shows packets that are:

- HTTP
 - AND from/to 192.168.1.10
-

◆ Example 2: SSH traffic to a server

```
tshark -i eth0 -Y "tcp.port == 22 && ip.dst == 192.168.1.20"
```

✓ Shows SSH packets going to that server

2□ OR Operator (||)

☞ **Meaning: Any one condition can be true**

◆ Example 1: HTTP or HTTPS traffic

```
tshark -i eth0 -Y "tcp.port == 80 || tcp.port == 443"
```

✓ Shows web traffic (HTTP **or** HTTPS)

◆ Example 2: TCP or UDP traffic

```
tshark -i eth0 -Y "tcp || udp"
```

✓ Shows packets using TCP **or** UDP

3□ NOT Operator (!)

☞ **Meaning: Remove / exclude packets**

◆ Example 1: Hide DNS traffic

```
tshark -i eth0 -Y "!dns"
```

✓ Shows everything **except** DNS

◆ Example 2: Exclude one IP

```
tshark -i eth0 -Y "!ip.addr == 192.168.1.5"
```

✓ Shows packets **not related** to that IP

4□ Using Brackets (Very Important 🔥)

Brackets control **order of logic**.

◆ Example: Web traffic from one IP

```
tshark -i eth0 -Y "ip.addr == 192.168.1.10 && (tcp.port == 80 || tcp.port == 443)"
```

✦ Means:

- Traffic from 192.168.1.10
 - AND (HTTP **or** HTTPS)
-

5□ Real-World Simple Use Cases

🔒 Find login requests (HTTP only)

```
tshark -i eth0 -Y "http.request.method == POST && ip.addr == 192.168.1.10"
```

✂ □ Detect possible SSH brute force

```
tshark -i eth0 -Y "tcp.port == 22 && tcp.flags.syn == 1"
```

🚫 Remove noisy traffic

```
tshark -r capture.pcap -Y "!(arp || dns)"
```

One-Page Cheat Sheet

Logic	Example
AND	ip.addr==X && http
OR	`tcp.port==80
NOT	!dns
Brackets	`ip && (80

🔑 Simple Summary

- Logical filters **combine display filters**
- && = AND (both true)
- || = OR (any one true)
- ! = NOT (exclude)
- Used with **-Y** in TShark